

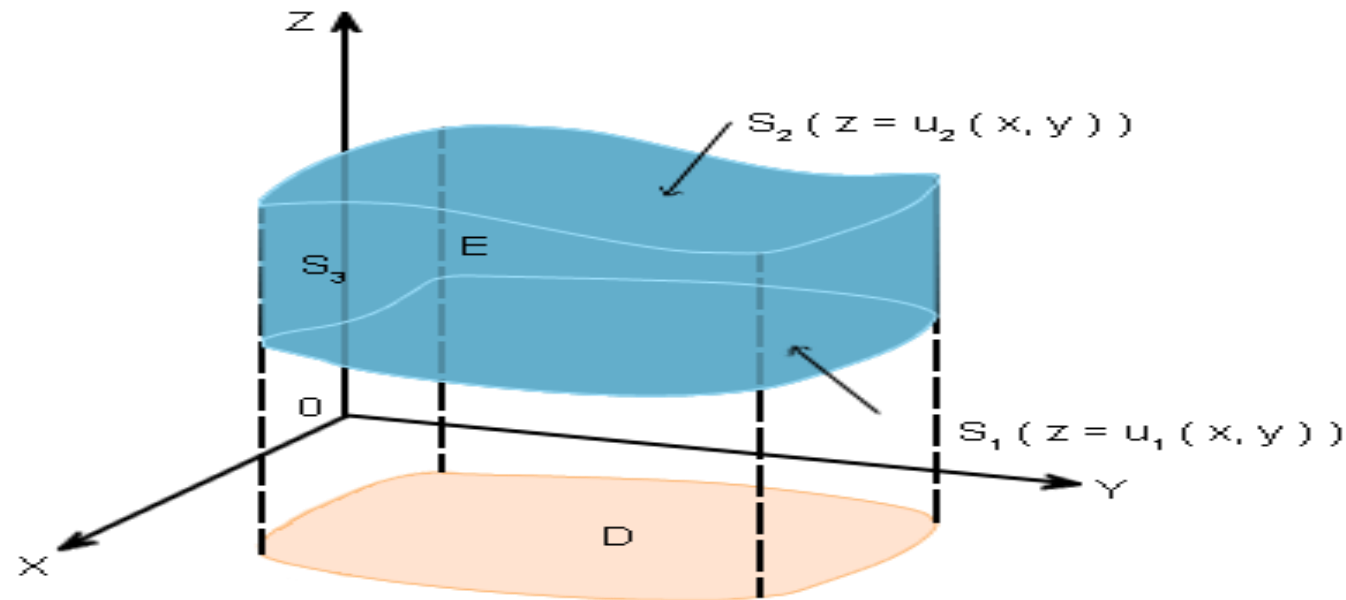
MTH 166

Lecture-37

Gauss's Divergence Theorem

Statement: Let D be a closed and bounded region in 3-dimensional space whose boundary is a piecewise smooth surface S oriented outwards. Let $\vec{V}$ be a vector which is continuous and have continuous first order partial derivatives. Let $\hat{n}$ is a unit normal vector drawn outward to the surface S . Then:

$$\iint_S (\vec{V} \cdot \hat{n}) dA = \iiint_D (\text{div } \vec{V}) dv = \iiint_D (\vec{\nabla} \cdot \vec{V}) dv$$



Important Results from MCQ point of view:

1. By gauss divergence theorem: $\iint_S (\vec{V} \cdot \hat{n}) dA = \iiint_D (\text{div } \vec{V}) dv = \iiint_D (\vec{\nabla} \cdot \vec{V}) dv$
2. Gauss Divergence theorem gives a relationship between double integral and triple integral, unlike Green's theorem and Stokes' theorem.
3. Gauss Divergence theorem is applicable to a closed region bounded by a surface, where as Stokes' theorem is applicable to an open surface bounded by a closed curve.
4. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ (a constant vector) and V is the volume,
(I). $\iint_S (\vec{a} \cdot \hat{n}) dA = 0$
(II). $\iint_S (\text{curl } \vec{r} \cdot \hat{n}) dA = 0$
(III). $\iint_S (\vec{r} \cdot \hat{n}) dA = 3V = 3(\text{Volume of the given region})$

Problem 1: Evaluate $\iint_S (\vec{V} \cdot \hat{n}) dA$ using Gauss Divergence Theorem, where $\vec{V} = x\hat{i} + y\hat{j} + z\hat{k}$ and D is the region bounded by the sphere $x^2 + y^2 + z^2 = 16$.

Solution: Here: $\vec{V} = x\hat{i} + y\hat{j} + z\hat{k}$

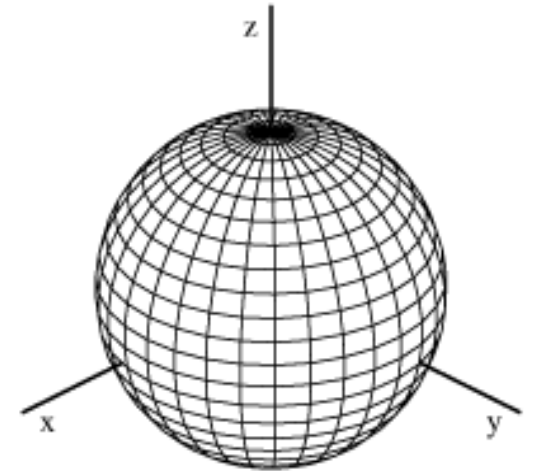
$$\Rightarrow \operatorname{div} \vec{V} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1$$

$$\Rightarrow \operatorname{div} \vec{V} = 3$$

By Gauss Divergence Theorem: $\iint_S (\vec{V} \cdot \hat{n}) dA = \iiint_D (\operatorname{div} \vec{V}) dv$

$$\Rightarrow \iint_S (\vec{V} \cdot \hat{n}) dA = \iiint_D 3 dv = 3 \iiint_D dv = 3(\text{Volume of sphere } x^2 + y^2 + z^2 = 16.)$$

$$\Rightarrow \iint_S (\vec{V} \cdot \hat{n}) dA = 3 \left(\frac{4}{3} \pi r^3 \right) = 4\pi(4^3) = 256\pi \text{ Answer.}$$



Problem 2: Evaluate $\iint_S (\vec{V} \cdot \hat{n}) dA$ using Gauss Divergence Theorem, where $\vec{V} = x\hat{i} + y\hat{j} + z\hat{k}$ and D is bounded by the edges $0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$.

Solution: Here: $\vec{V} = x\hat{i} + y\hat{j} + z\hat{k}$

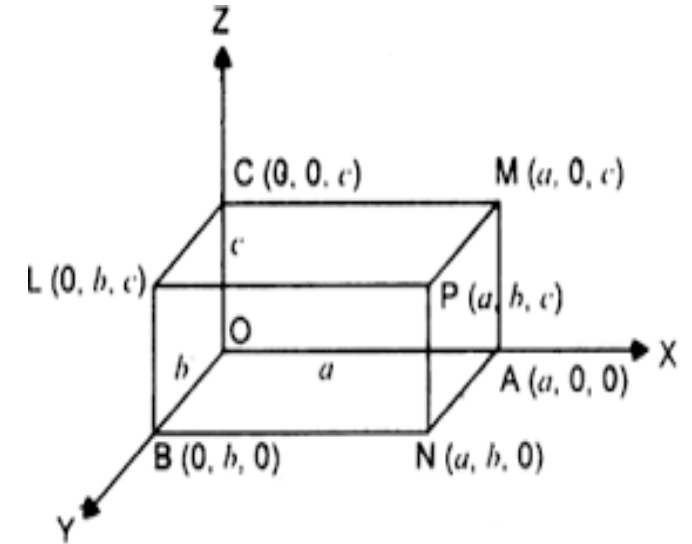
$$\Rightarrow \operatorname{div} \vec{V} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1$$

$$\Rightarrow \operatorname{div} \vec{V} = 3$$

By Gauss Divergence Theorem: $\iint_S (\vec{V} \cdot \hat{n}) dA = \iiint_D (\operatorname{div} \vec{V}) dv$

$$\Rightarrow \iint_S (\vec{V} \cdot \hat{n}) dA = \iiint_D 3 dv = 3 \iiint_D dv = 3(\text{Volume of cuboid.})$$

$$\Rightarrow \iint_S (\vec{V} \cdot \hat{n}) dA = 3(lbh) = 3(abc) \text{ Answer.}$$



Problem 3: Evaluate $\iint_S (\vec{V} \cdot \hat{n}) dA$ using Gauss Divergence Theorem,

where $\vec{V} = 3x^2\hat{i} + 6y^2\hat{j} + z\hat{k}$ and D is the region bounded by the closed cylinder $x^2 + y^2 = 16$, $z = 0$ and $z = 4$.

Solution: Here: $\vec{V} = 3x^2\hat{i} + 6y^2\hat{j} + z\hat{k}$

$$\Rightarrow \operatorname{div} \vec{V} = \frac{\partial}{\partial x}(3x^2) + \frac{\partial}{\partial y}(6y^2) + \frac{\partial}{\partial z}(z)$$

$$\Rightarrow \operatorname{div} \vec{V} = 6x + 12y + 1$$

By Gauss Divergence Theorem: $\iint_S (\vec{V} \cdot \hat{n}) dA = \iiint_D (\operatorname{div} \vec{V}) dv$

$$\Rightarrow \iint_S (\vec{V} \cdot \hat{n}) dA = \iiint_D (6x + 12y + 1) dv$$

Let us calculate the limits of x, y, z

$$0 \leq z \leq 4 \text{ (Given height of cylinder)}$$

$$\text{Also } x^2 + y^2 = 16$$

$$\Rightarrow y = \pm\sqrt{16 - x^2}$$

$$\Rightarrow -\sqrt{16 - x^2} \leq y \leq \sqrt{16 - x^2}$$

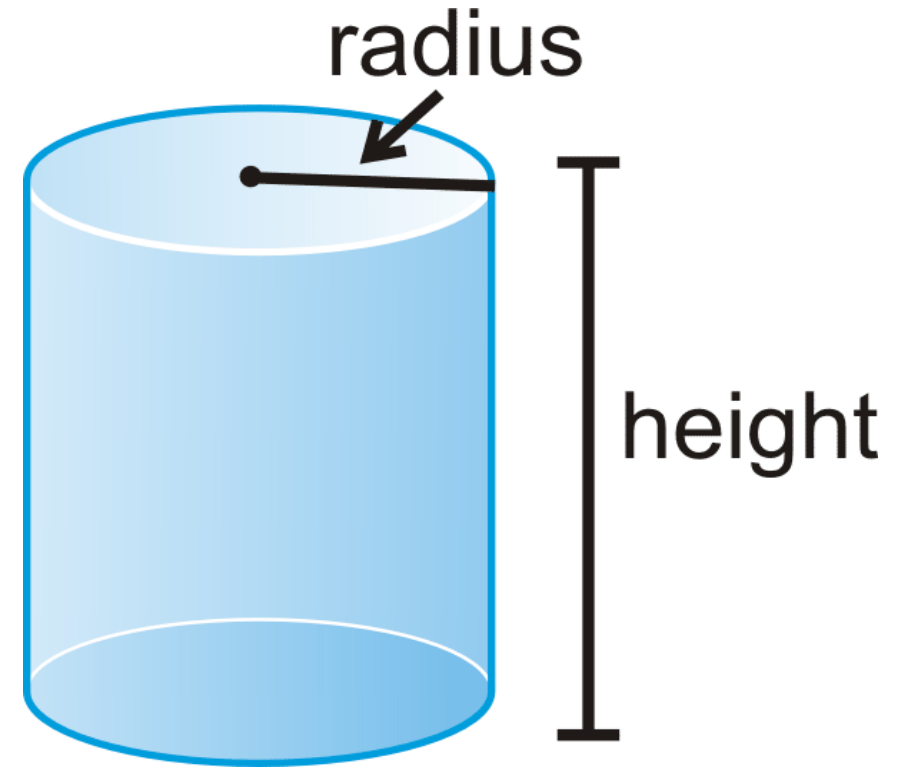
On x-axis (y=0)

$$\Rightarrow x^2 + 0^2 = 16$$

$$\Rightarrow x = \pm 4$$

$$\Rightarrow -4 \leq x \leq 4$$

Let us put these limits in the given integral



$$\Rightarrow \iint_S (\vec{V} \cdot \hat{n}) dA = \iiint_D (6x + 12y + 1) dv$$

$$\Rightarrow \iint_S (\vec{V} \cdot \hat{n}) dA = \int_{z=0}^{z=4} \int_{x=-4}^{x=4} \int_{y=-\sqrt{16-x^2}}^{\sqrt{16-x^2}} (6x + 12y + 1) dy dx dz$$

$$= \int_{z=0}^{z=4} dz \int_{x=-4}^{x=4} \int_{y=-\sqrt{16-x^2}}^{\sqrt{16-x^2}} (6x + 12y + 1) dy dx$$

$$= 4 \int_{x=-4}^{x=4} \int_{y=-\sqrt{16-x^2}}^{\sqrt{16-x^2}} (6x + 12y + 1) dy dx$$

$$= 4 \int_{x=-4}^{x=4} \int_{y=-\sqrt{16-x^2}}^{\sqrt{16-x^2}} (1) dy dx \text{ (Because } x \text{ and } y \text{ are odd functions)}$$

$$= 4(2)(2) \int_{x=0}^{x=4} \int_{y=0}^{\sqrt{16-x^2}} (1) dy dx \text{ (Because } 1 \text{ is an even function)}$$

$$= 16 \int_{x=0}^4 \sqrt{16-x^2} dx = 16 \left[\frac{x\sqrt{16-x^2}}{2} + \frac{16}{2} \sin^{-1} \left(\frac{x}{4} \right) \right]_{x=0}^4$$

$$= 64\pi \text{ Answer.}$$



Thanks!